

CHECK WHAT YOU LEARNT

Check Yourself - Unit 3**A. Choose the correct answer**

- The traffic module needs only four wires because
(a) it is small (b) resistors are built in (c) it has no lamps (d) it uses no power
- If the GND wire is removed, the module will
(a) get brighter (b) not work at all (c) change colour (d) run faster
- delay(5000) in the red section means red stays on for
(a) 5 milliseconds (b) 50 seconds (c) 5 seconds (d) 5 minutes
- A flowchart should be drawn
(a) after coding (b) before coding (c) instead of coding (d) never
- To make green last longer you would change
(a) the wire colour (b) the delay value after green (c) pinMode (d) the GND pin
- Yellow exists on a real signal so that drivers can
(a) speed up (b) slow down safely (c) see better (d) turn left

B. Fill in the blanks

- The three lamps are connected to pins _____, _____ and _____.
- Everything inside void loop() repeats _____.
- A flowchart shows the same idea as code, but in _____.
- Late at night a real signal may switch to _____ mode.

C. Answer in one or two lines

- Why is the yellow lamp needed? What would go wrong on a real road without it?
- Write down the delay values you chose and explain why you chose those numbers.
- Describe one change you made after testing, and what made you change it.
- A classmate's module does nothing at all. Name the two wires you would check first.

UNIT 4

Intermediate Project: Automatic Night Lamp

Sessions 9, 10 and 11

Until now your machine has been blind. It followed orders and nothing more. In this unit it gains its first sense, and for the very first time it makes a decision of its own.

By the end of this unit you will be able to:

- Explain how an LDR turns light into a number.
- Read a live sensor value on the Serial Monitor.
- Choose a threshold from real measurements instead of guessing.
- Write an if - else decision that links an input to an output.

SESSION 9

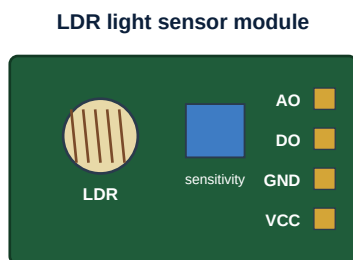
one class

A Sensor That Feels Light

Today you will

- Learn how an LDR works and what its numbers mean.
- Open the Serial Monitor and watch a live sensor value.
- Record the reading in bright light and in darkness.

An LDR - a Light Dependent Resistor - is a small disc with a wiggly line printed across it. In bright light, electricity passes through that wiggly line easily. In darkness, it resists. The Arduino cannot see brightness the way you can. It can only read that resistance as a **number between 0 and 1023**, and it is up to you to decide what those numbers mean.

**What the numbers mean****Bright light**

resistance drops

-> small number on A0

Darkness

resistance rises

-> big number on A0

The Arduino cannot 'see'. It only reads a number from 0 to 1023, and you decide which number counts as dark.

Figure 4.1 - The LDR module and the numbers it gives you

This is one of the biggest ideas in the whole book, so it is worth saying slowly. A sensor does not tell a machine *it is dark*. A sensor gives a number. A human decides which numbers count as dark. That decision is called setting a **threshold**, and choosing it well is the difference between a lamp that works and a lamp that flickers all evening.

Try This

Wire the LDR module: VCC to 5V, GND to GND, AO to A0. Then upload a program with only these two lines inside loop(): `Serial.println(analogRead(A0));` and `delay(200);` - and remember `Serial.begin(9600);` inside setup(). Open Tools > Serial Monitor and cover the sensor with your hand. Write down the number in a bright room and the number under your palm. Those two numbers decide everything that follows.

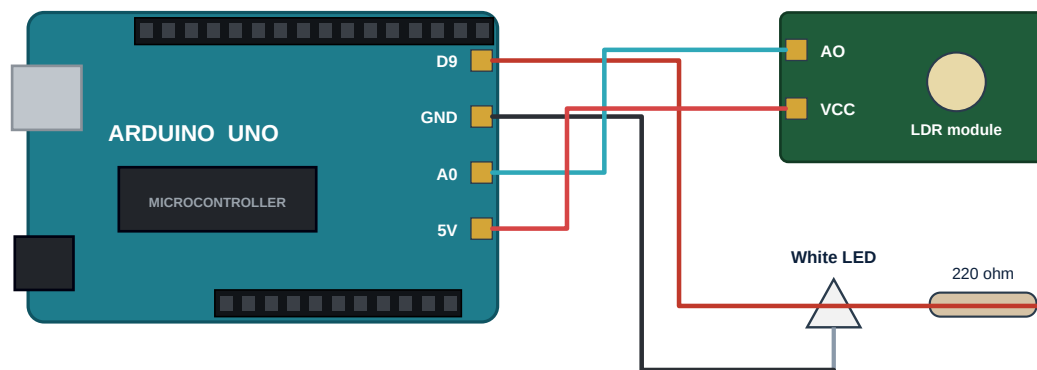
SESSION 10

one class

Building the Night Lamp**Today you will**

- Wire the LDR module and the white LED together.
- Understand the if - else decision.
- Upload the night lamp program and watch it decide.

Automatic night lamp - connections



Sensor goes to the analog side (A0). The lamp goes to the digital side (D9).

Figure 4.2 - Input on one side, output on the other

Component pin	Arduino pin	Why
LDR module VCC	5V	Power for the sensor
LDR module GND	GND	Shared return path
LDR module AO	A0	The light level, as a number
White LED long leg	D9 through a 220 ohm resistor	The street lamp
White LED short leg	GND	Return path

Now for the new idea. Until this moment your programs have only given orders. This one asks a question first. In plain English: *if it is dark, switch the lamp on; otherwise switch it off.* In Arduino that is an **if - else**, and in a flowchart a question is drawn as a diamond rather than a box.

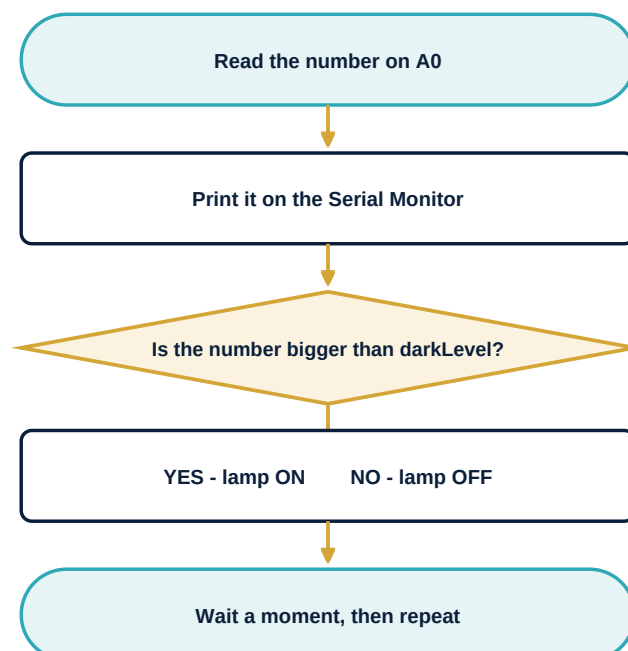


Figure 4.1 - A decision is drawn as a diamond

```

NightLamp.ino
int ldrPin    = A0;
int lampPin   = 9;
  
```

```

int darkLevel = 600;    // change this after your own testing

void setup() {
  pinMode(lampPin, OUTPUT);
  Serial.begin(9600);    // start talking to the laptop
}

void loop() {
  int light = analogRead(ldrPin);    // a number from 0 to 1023
  Serial.println(light);             // show it on the Serial Monitor

  if (light > darkLevel) {
    digitalWrite(lampPin, HIGH);    // it is dark - lamp ON
  } else {
    digitalWrite(lampPin, LOW);     // bright enough - lamp OFF
  }

  delay(200);
}

```

Watch Out

600 is only a starting guess, and our classroom is not your classroom. If your lamp behaves backwards - on in daylight and off in the dark - do not rewire anything. Simply change $>$ to $<$. Some LDR modules count upwards in darkness and others count downwards.

SESSION 11

one class

Calibration: Finding Your Own Number**Today you will**

- Measure real readings in your own classroom.
- Choose and justify your threshold value.
- Improve the lamp so it does not flicker at dusk.

Calibration means tuning a machine to the real world it has to live in. A thermometer is calibrated against boiling water. A weighing scale is calibrated against a known weight. Your night lamp is calibrated against your own classroom, at the time of day you actually use it.

Condition	Number on Serial Monitor	Lamp should be
Bright classroom, tube lights on	_____	OFF
Lights off, curtains drawn	_____	ON
Hand fully covering the sensor	_____	ON
Sensor near the window at 5 pm	_____	your decision
The darkLevel you chose	_____	roughly halfway between

There is one more problem to solve. At dusk the light level sits almost exactly on your threshold, and tiny changes make the lamp flicker on and off repeatedly. Real engineers fix this by making the machine slightly stubborn: once the lamp is on, it needs the light to rise well above the threshold before it will switch off again. Try adding a `delay(1000)` after each switch and see how much steadier the lamp becomes.

Challenge

Real street lamps do not simply jump from off to full brightness. Add a second LED so the lamp glows dim at dusk and bright in full darkness, using `analogWrite(pin, value)` where value runs from 0 to 255. Hint: only the pins marked with ~ can do this.

New word	What it means
Sensor	A part that turns something physical into a number.
Analog	A value anywhere in a range, not just on or off.
Threshold	The number at which a decision changes from one answer to the other.
Calibration	Adjusting a machine to suit the real place it works in.